1.	**Prerequisites**	A knowledge of the specifications for P1, P2, P3, P4 and M1, and their prerequisites and associated formulae, is assumed and may be tested.
2.	**Examination**	• First assessment: June 2020. • The assessment is 1 hour and 30 minutes. • The assessment is out of 75 marks. • Students must answer all questions. • Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*. • The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
3.	**Notation and formulae**	Students will be expected to understand the symbols outlined in *Appendix 7: Notation*. Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage. Kinetic energy = $\frac{1}{2}mv^2$ Potential energy = mgh

M2.3 Unit content

What students need to learn:		Guidance
1. Kinematics of a particle moving in a straight line or plane		
1.1	Motion in a vertical plane with constant acceleration, e.g. under gravity.	
1.2	Simple cases of motion of a projectile.	
1.3	Velocity and acceleration when the displacement is a function of time.	The setting up and solution of equations of the form $\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{f}(t)$ or $\frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{g}(t)$ will be consistent with the level of calculus in P1, P2 P3 and P4.
1.4	Differentiation and integration of a vector with respect to time.	For example, given that $\mathbf{r} = t^2\mathbf{i} + t^{\frac{3}{2}}\mathbf{j}$, find $\dot{\mathbf{r}}$ and $\ddot{\mathbf{r}}$ at a given time.